Anatomy from practical perspective :→

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① Synovial Joint - Diagram
- Classification
- Example

② Classification of fibrous joint

③ Sesamoid Bone

④ Cartilageous Joint - Classification
- Example

⑤ Pivot Joint

⑥ Blood supply of long bone ⑦ Epiphysis

★ Synovial joint ⇒ Diarthrosis

① Any most evolved, most mobile type of synovial joint

Classification :→

① Plane / Gliding type → Gliding movement ⟷ Hand

② Uniaxial

Transverse A. • Hing Joint → flexion & Extension

Vertical Axis • Pivot Joint → Rotation

③ Biaxial

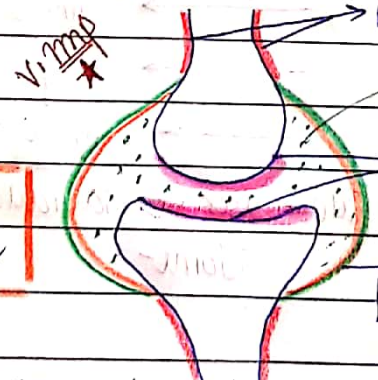
• Condylar → TMJ, knee

• Ellipsoid → wrist

④ Multiaxial

• Saddle joint →

• Ball & Socket joint →



orange ⇒ synovial membrane

• → b/w two articulating surfaces partition called menisci may be present or not.

• → Articular capsule is lined by synovial membrane which secretes synovial fluid

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Watch -Dog Action

• → Due to fibrous nature of articular capsule it is sensitive to sprain by appropriate reflexes (helpful mechanism)

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• → Varying degree of movements are permitted by synovial joint-

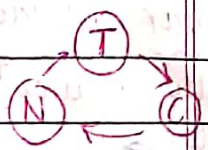
Classification →

Upper limb

- Sternoclavicular → Saddle
- Acromio clavicular → Plane
- ★ Shoulder / Glenohumeral → Ball & Socket-
- ★ Elbow → Hinge
- Radio ulnar
  - → Sup. → Pivot
  - → Middle → Syndesmosis (fibrous)
  - → Inf. → Pivot-
- ★ Wrist / Radiocarpal → Ellipsoid
- Carpometacarpal → Saddle
- MCP joint → Ellipsoid
- Inter-phalangeal J → Hinge

Lower limb →

- Hip Joint → Ball and Socket-
- Knee Joint → f & T → Condylar joint-
  - f & Patella → Saddle joint-
- Tibiofibular Joint → Sup → Plane joint-
  - → Inf. → Syndesmosis (fibrous)
- Ankle → Hinge
- MTP joint → Ellipsoid
- Interphalangeal → Hinge



⇒ Ball & Socket- ⇒ Talo-calceno-Navicular Joint-

⑥ Plane →
1. Chondro-sternal (except 1st)
2. costo-vertebral
3. Acromio-clavicular joint-

Imagine = 4. Superio-tibio-fibular

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Thorax → Chondrosternal joint → Plane

★★ but 1st chondrosternal joint is → fibrous ★★★ (unusual)

Costo-vertebral joint → Plane

Ear-ossicles → Incudo-stapedial → Ball & Socket

Incudo-malleolar → Saddle

PAA → Atlanto-Axial ⇒ Pivot joint

→ Atlanto-occipital ⇒ Ellipsoid

NOTES → ① Ball & Socket → ① Shoulder joint

② Hip joint

★③ Talo-calcaneo-Navicular joint

★④ Incudo-stapedial joint.

② Hinge → ① Elbow

② Ankle

③ Interphalangeal

③ Condylar → ① Knee (F-T)

★② TMJ

④ Ellipsoid → ① MCP

② MTP

③ Wrist

④ Atlanto-occipital

⑤ Saddle → ① Incudo-malleal joint ★

② Sterno-clavicular joint

③ Carpo-metacarpal joint

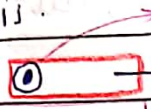
④ Knee (F&P) joint

⑤ Calcaneo-cuboid joint

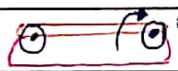
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⑦ Pivot → ① Atlantoaxial joint.
② Sup. & Inf. radioulnar joint.

① Plane → दो bones जो articulate होती are flat
→ promote gliding movement in various directions (see on net)

② Hinge → • Articular surfaces → pulley shaped
• Movement permitted किस plane में around transverse axis.
→ Axis जो image में आ रही है paper को (transverse)
Eg → 

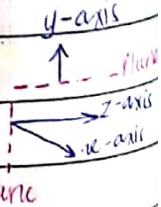
(flexion, Extension)
उसी में movement only permitted up & down
No other movement so it is Hinge joint.
→ same is case of Ankle | remember side to side movement जो at ankle is of tibio-fibular joint-Sup which is of plane variety] ***

③ Pivot →
→ • जैसे hinge में plane था transverse here plane is vertical


(superior view) here ulna will be fixed radius will move as shown by arrow.

** Conclusion ① → Hinge & Pivot both have uniaxial movement
② → Hinge ⇒ Transverse plane ⇒ up & down mov.
Pivot ⇒ Vertical plane ⇒ Rt & Lt mov.

④ Condylar → Among two articulating surfaces
1 → convex
2 → concave
→ attached by ligament
→ Rather than biaxial it should be biplanar
1 plane ⇒ around transverse axis
1 plane ⇒ around vertical axis
Eg → ① TMJ
② Knee joint



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⑤ Ellipsoid → Among two articulating surfaces

↓ → convex

↓ → concave

→ bi-axial → around ~~Transverse~~ axis

→ around ~~Antero-posterior~~ axis.

** Eg → wrist joint

Conclusion ① → Condylar & Ellipsoid have biaxial movements

→ होते हैं saddle हैं ⑥

⑥ Saddle → Almost same as ellipsoid but movements around all three axis

⑦ Ball & Socket →

- Globular head fitting in cup-shaped socket
- Shoulder, hip Joint

V. imp *

(M) ⇊ Saddle (I) ⇊ (S) Ball & socket

★ fibrous joint : → Sutures, Gomphosis, Syndesmosis

→ bones articulated by fibrous tissue

→ either → immovable
→ or slight movement

Types → ① Sutures ⇒

* present only in skulls.
* Sutural side of each bone covered with layer of osteogenic cells

Types ⇒

① Plane → Sutural ligament
→ osteogenic cells

Ex Internasal, Interpalatine.

② Squamous → overlapping bones

Ex Tempro-parietal suture.

③ Serrate →

Ex Sagittal suture

④ Denticulate → (Strongest) ★★

Ex Lambdoidal suture (λ)

⑤ Schindylesis → → fastnum of sphenoid fitting in → Ala of vomer bone

② Gomphosis →

→ peg & socket junction
→ connected via periodontal ligament

Ex connection b/w tooth & its socket

③ Syndesmosis →

→ Joined by interosseous membrane
↳ b/w bone.

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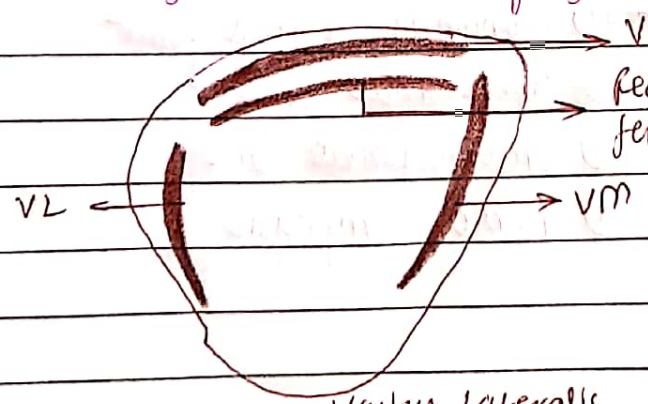
Ex → (1) Middle radio-ulnar joint
(2) Inferior tibio-fibular joint
(3) Sacro-iliac joint

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NOTE ** → In neonates skull two sutures i.e sutures are joined by fontanelles (Temporary) → 6 होते हैं
→ These allow underlying bone to increase their size
→ Anterior fontanelle ⇒ judge hydration of baby.
→ fontanelle judge intra-membranous ossification in progress.

★ Sesamoid bone ★

→ bony masses found in tendons or joint capsules
→ have no periosteum
→ can be present at articular or non-articular surface → periosteum.
→ covered by Hyaline cartilage
→ lubricated by synovial membrane.
Ex → Patella in tendon of Quadriceps femoris
Pisiform in tendon of flexor carpi ulnaris



Quadriceps femoris → Vastus Lateralis
→ Vastus Intermedius
→ Vastus medialis
→ Rectus femoris

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★ Cartilagenous Joint ★

→ Joined by cartilage in b/w

Type → ① Primary ⇒ (TEMPORARY)

Syn-chondrosis



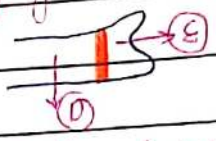
↓

Hyaline cartilage

eventually with age the cartilage will ossify converting

Syn-chondrosis → to → syn-ostosis

Ex → ① b/w Epiphysis & diaphysis of long bone

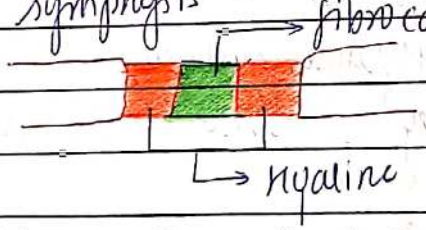


② costo-chondral joint

③ spheno-occipital joint

② secondary ⇒ (PERMANENT)

Symphysis



→ as fibro-cartilage don't ossify this joint is permanent

Ex → ① Manubrio-sternal joint

② Xiphisternal joint

③ Intervertebral disc

④ Pubic symphysis

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★ Blood supply of Long bone ★

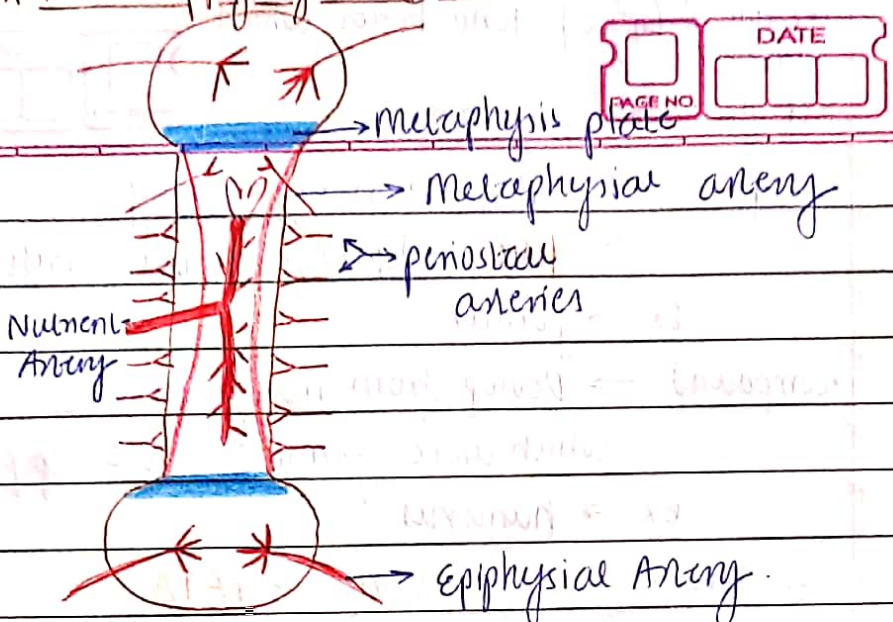


fig. blood supply of long bone

A] Arterial supply :→

ⓐ Nutrient artery. ⇒ (Inner 2/3rd cortex)

① Enters shaft through nutrient foramina

② Runs through cortex & divides into ascending & descending branches in medullary cavity.

③ Each branches, divides & redivides

④ Terminates by anastomosing with metaphysial & periosteal arteries.

ⓑ Periosteal artery ⇒

① Numerous, below cartilage & ligaments.

② Supply outer 1/3rd cortex

ⓒ Epiphysial artery ⇒

① Most of foramina here are for venous exit

ⓓ Metaphysial artery ⇒

① Derived from neighbouring systemic vessels.

NOTE → Direction of nutrient foramina away from growing ends

→ Growing ends → Towards the elbow I go from the knee I flee.

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★ Epiphysis ★

→ Ends & tips of long bones which later ossify

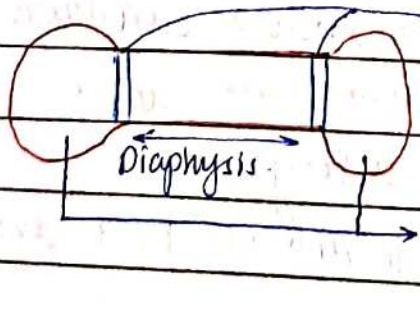
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Classification → (1) According to number →

- Simple → Develop from many epiphysis this fuse independently with shaft
  Ex → femur
- Compound → Develop from many centres which unite forming single epiphysis
  Ex → humerus

(2) According to function → PATA

- Pressure → involved in weight transmission
  Ex → Head of humerus
  Lower end of radius
- Atavistic → Evolutionarily अलग bone होती है fuse with another bone
  Ex → Coracoid process of scapula
- Traction → Not involved in weight transmission
  Ex → Tubercle of humerus
  Trochanter of femur
- Aberrant → Not always present
  Ex → b/w 1st & 2nd Metacarpal bone



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★ Clinicals : →

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① Suffix = itis
means inflammation

eg → Appendicitis
→ Tonsillitis
→ Arthritis (Inflammation of joints)
→ Neuritis
→ Dermatitis

② Suffix = ectomy
mean body से निकालना देना

Example → Appendicectomy
→ Tonsillectomy
→ Gastrectomy
→ Nephrectomy

③ Suffix = otomy
Incision करना body के hollow organ को open
किया then इसको close करके दिया after diagnosis.

Example → Laparotomy → (Lapr = Abdominal wall)
→ hysterotomy → (Hyster = Uterus)
→ cystotomy → (Incision in urinary bladder)
→ cysto-litho-tomy → (Removal of bladder के Stones)

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④ Suffix = ostomy
शरीर कोई organ को open करके इसको खुला छोड़ देना
Example → cystostomy (for urinary bladder)
→ Colostomy (for large intestine)
→ Tracheostomy (गले वाला साइड के for respiration & food)

⑤ Suffix = oma
means tumor
Example → Lipoma (lump under skin)
→ Osteoma (bone tumor)
→ Neurofibroma (Nerve sheath tumor in PNS)
→ haemangioma (blood vessels lump under skin)
→ Carcinoma (cancerous tumor)

⑥ Puberty → Age at which 2° sexual character appear.

⑦ Symptoms → complaints of a patient

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(8) Signs → findings of a doctor

(9) Diagnosis → Identification of disease, or determination of nature of disease

(10) Prognosis → Tracings जिसमें से Disease हुई

(11) Pyrexia → fever

(12) Lesion → Injury, pathological change in tissue

(13) Inflammation → Local reaction of tissues to injury by physical, chemical, biological agent.

* Characterised by → (a) Swelling
(b) Pain
(c) Redness
(d) Warmth or heat
(e) Loss of function.

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(14) Oedema → Swelling due to accumulation of fluid in ECF

(15) Thrombosis → Intra-vascular coagulation (solidification) of blood

(16) Embolism → Occlusion of vessel by detached & circulating Thrombus

(17) Haemorrhage → Bleeding which can be Internal or External

(18) Ulcers → Gap/erosions in the surface continuity of skin or mucous membrane

(19) Sinus → A blind track open at one end lined by epithelium.

(20) Fistula → Abnormal passage b/w two internal organ to organ
→ or organ to surface of body
Lined by epithelium

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(21) Necrosis → local death of tissue or organ due to irreversible damage to nucleus

(22) Degeneration → Retro-gressive change causing deviation in the structural & functional qualities
* Reversible
* may end in necrosis

(23) Gangrene → Necrosis + Putrefaction → Decay, Rotting.

(24) Infarction → Necrosis of tissue due to sudden obstruction of its artery of supply

(25) Atrophy → Diminishing size of cells, tissues, organs or a part due to loss of nutrition

(26) Dystrophy → Diminishing size due to lack of nutrition
Ex → Muscular dystrophy.

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(27) Hypertrophy → Increase in size without increase in no. of cells

(28) Hyperplasia → Increase in size due to increased number of cells

(29) Hypoplasia → Incomplete development

(30) Aplasia → Failure of development

(31) Syndrome → A group of signs & symptoms constituting together the picture of disease

(32) Paralysis → Loss of motor power (movement) of a part of body due to denervation

(33) Hemiplagia → Paralysis of $\frac{1}{2}$ of the body

(34) Paraplagia → Paralysis of both lower limbs.

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(35) Monoplagia → Paralysis of both upper limbs

(36) Quadriplegia → Paralysis of all four limbs.

(37) Anaesthesia → Loss of sensation

(38) Analgesia → Loss of pain sensibility.

(39) Therm-Anaesthesia → Loss of Tempt. sensibility.

(40) Hyper-aesthesia → Abnormal increased sensibility.

(41) Par-aesthesia → Perverted (विकृत हुई) feeling of sensations.

(42) Coma → Deep unconsiousness

(43) Tumor (neoplasm) → A (circumscribed → limited to that area), non-inflammatory, Abnormal growth arising from body tissues

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(44) Benign → mild illness or growth which don't endanger life

(45) Malignant → Severe illness or growth which is resistant to treatment

(46) Carcinoma → Malignant growth arising from epithelium (ectoderm & endoderm)

(47) Sarcoma → Malignant growth arising from connective tissue

(48) Cancer → Any malignant neoplasm (Tumor)
* Invasive
* Can result in death if not treated well

(49) Metastasis → Spread of local disease to distant part of body.

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(50) Convalescence → Recovery period b/w end of disease & restoration to complete health.

(51) Therapy → Treatment of disease

* → Clinicals of Skeleton : →

(1) Cleido-cranial (Fair) dysostosis → Disorder of bone ossification. → Defect in membranous → mesodermal ossification

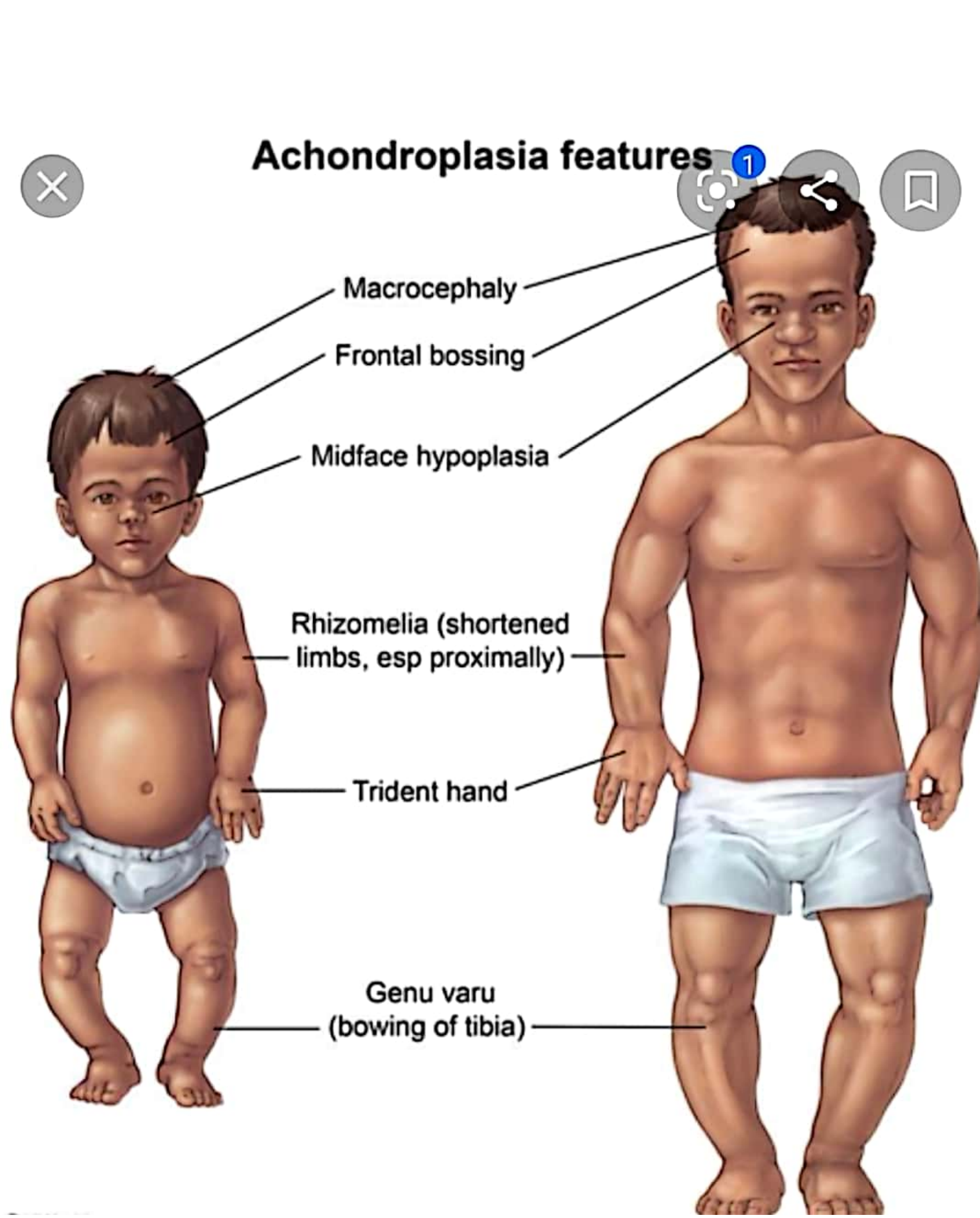
features → (i) Varying degree of Aplasia → not well developed of clavicles
(ii) Increase in transverse diameter of cranium
(iii) Retardation in fontanelle ossification
may be hereditary or environmental

(2) Achondroplasia → formation
no-cartilaceous ossification

* common type of dwarfism due to defect in endochondral ossification / cartilageous ossification.
* Limb → short, Trunk → normal
* Transferred in generations as Mendelian dominant character

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Achondroplasia features



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③ Periosteum → sensitive to tearing or tension.

→ Drilling in compact bone → Mild pain
without Anaesthesia ↳ Hard in nature

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→ Drilling in spongy bone → much more painful
spongy.

★ → As periosteum is supplied by somatic nerves
* fractures
* Tumors → of bone are painfull.
* Injections

④ Blood supply of bone is very rich so it is difficult to interrupt it & kill the bone =

★ → passing metal pin hardly interfares with its blood supply

⑤ Fracture →

2 types → simple → not connected with skin.
* can be → (a) Spiral
(b) Horizontal
(c) Oblique

→ compound → connected with skin
(open)

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Q. Healing of bone ?

Ans Occurs in ③ stages

→ Repair by granulation tissue

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→ Union by callus

→ Replaced by mature bone.

Conclusion → (Hematoma) → (callus) → (Matured bone)

Q. How hanging causes death ?

Ans In Hanging Axis (C_2) vertebrae gets fractured

→ if dens ★ gets fracture

[C_2 vertebrae का दाँत टूटना]

↓

it will hit important M.O centres

↳ Medulla Oblangata

↓

causing instant death.

⑥ → Rickets → caused due to vitamin-D deficiency.

* calcification of cartilage fails
* ossification of growth zone disturbed

→ Rickets affect developing bone so it affects the period जब max. growth होती है।

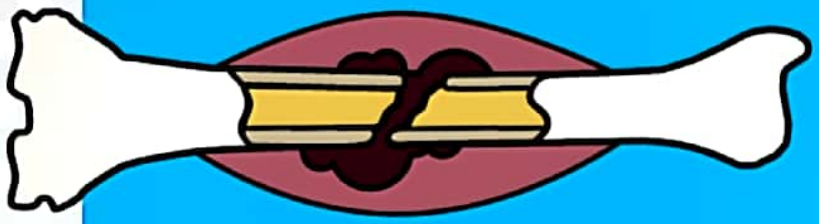
[3 months to 3 years]

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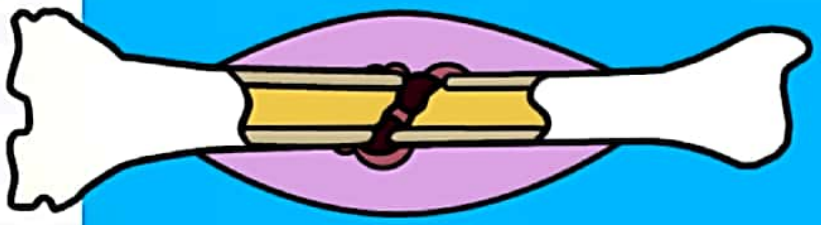
1. Hematoma Formation



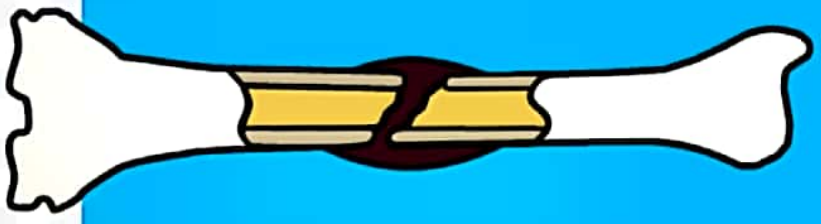
2. Granulation Tissue Formation (Inflammation)



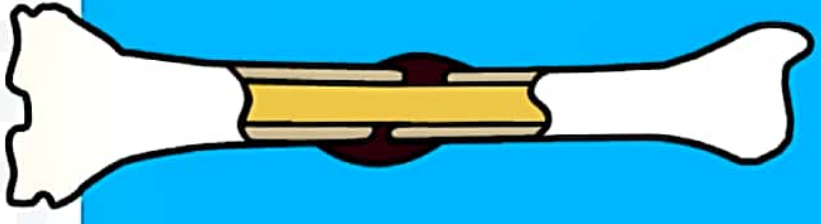
3. Callus Formation



4. Consolidation



5. Bone remodelling



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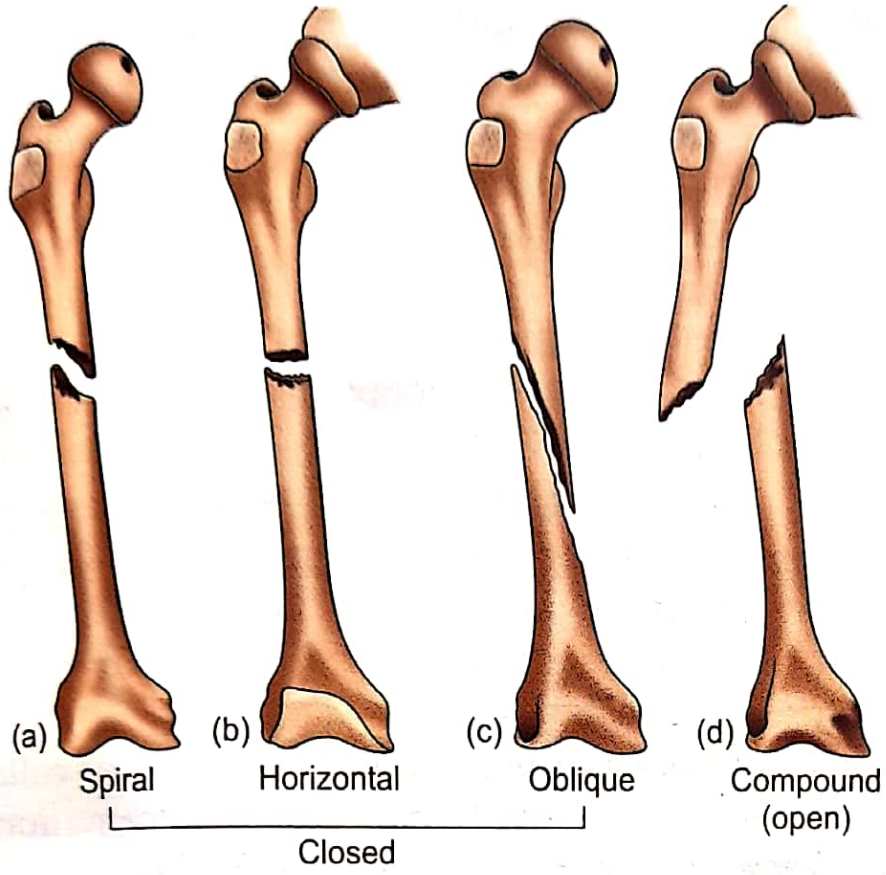


Fig. 2.26: Types of fractures

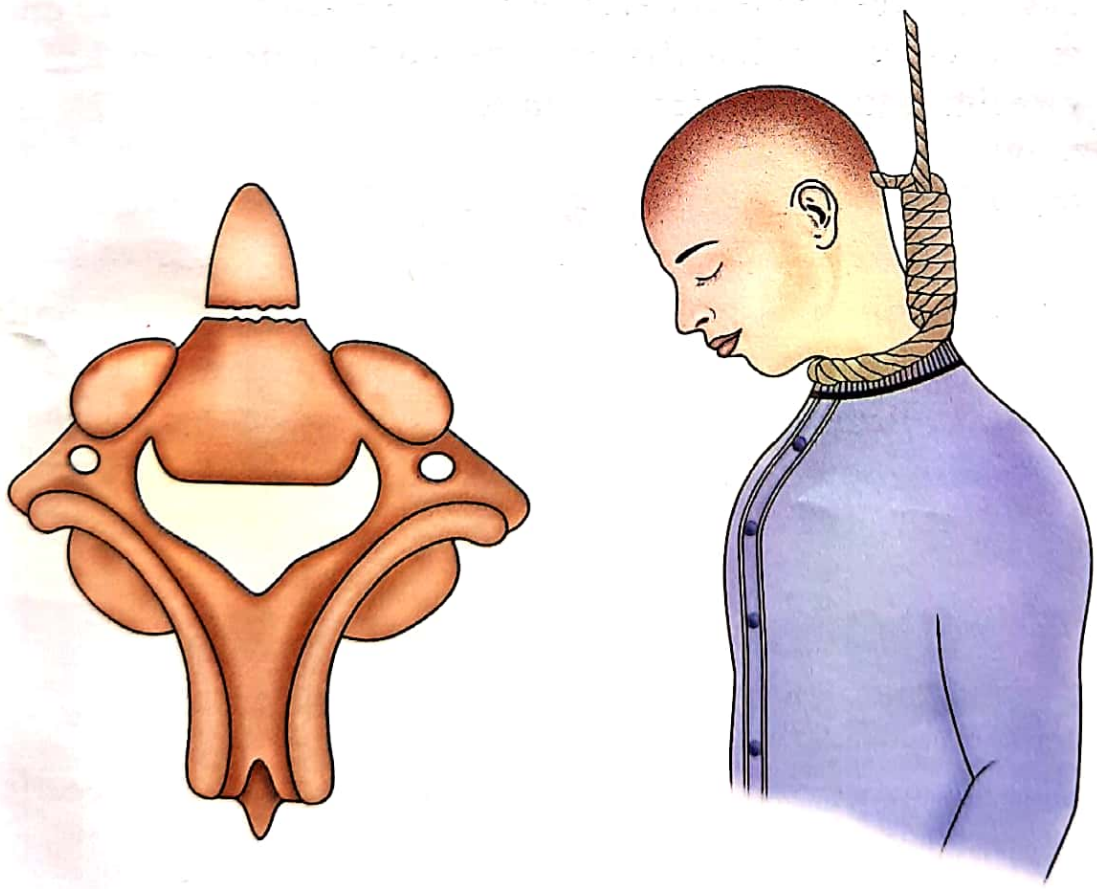


Fig. 2.27: Hanging leading to fracture of dens of axis vertebra

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① → Mineralization of bone doesn't take place.

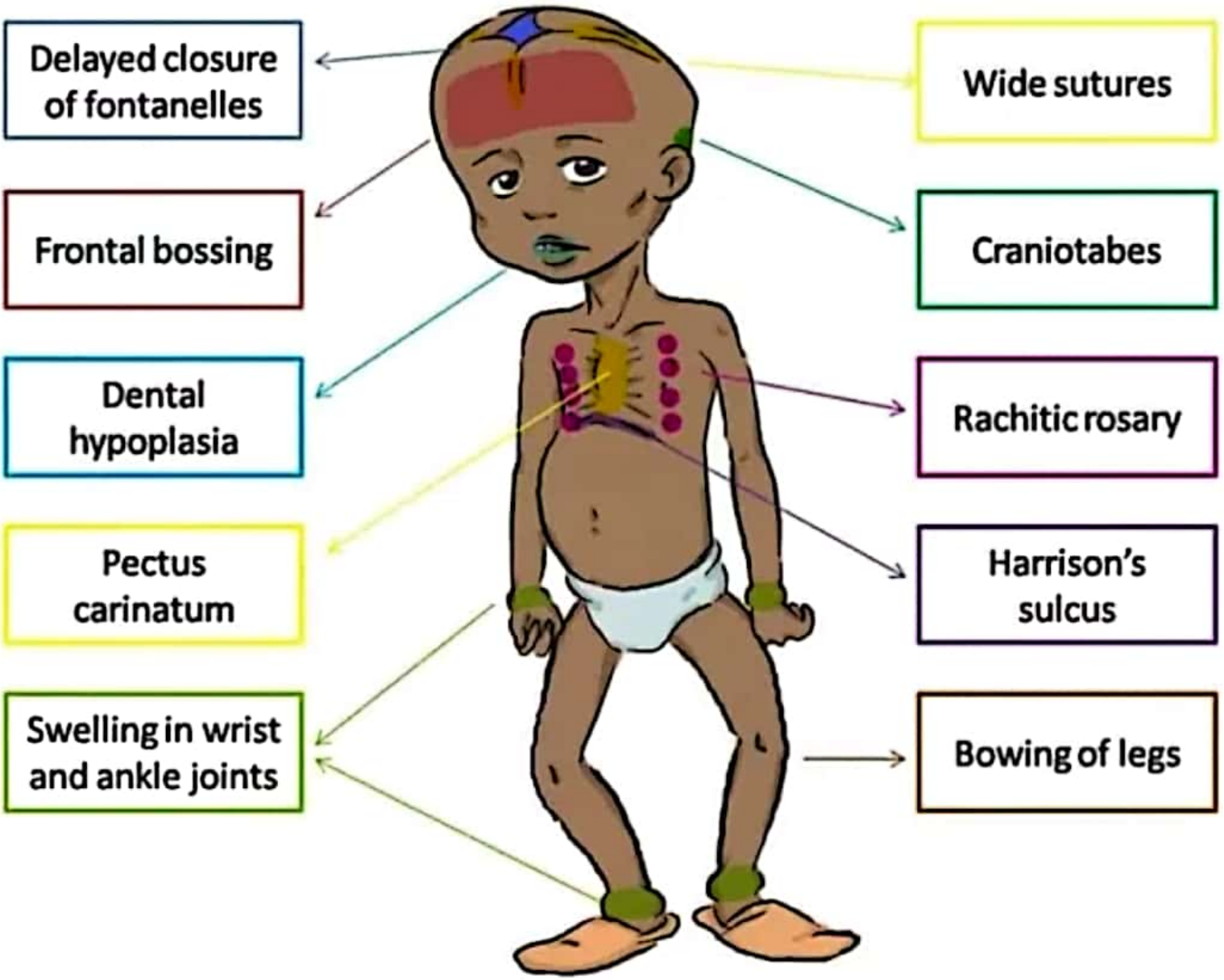
Result → ⓐ Craniotabes → softening or thinning of skull
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Expansion of Anterior Ribs ⇓ more prominent
ⓑ Rachitic rosary → at costochondral junction
ⓒ Harrison's sulcus → at Diaphragmatic attachment → sternum bone
ⓓ enlarged epiphysis in limb bones
ⓔ Pelvic & Spinal deformities.
जैसे bowing legs

② → Things required for development of bone
* Adequate proteins → Inadequate causes Rickets.
* Calcium
* Vitamin D.

③ → Scurvy → caused due to vitamin-C deficiency.
→ formation of Collagen fibres & Matrix impaired
→ as collagen का ↓ = rupture of capillaries & defective formation of new capillaries
→ Haematoma → in muscles & bones
causes pain & tenderness

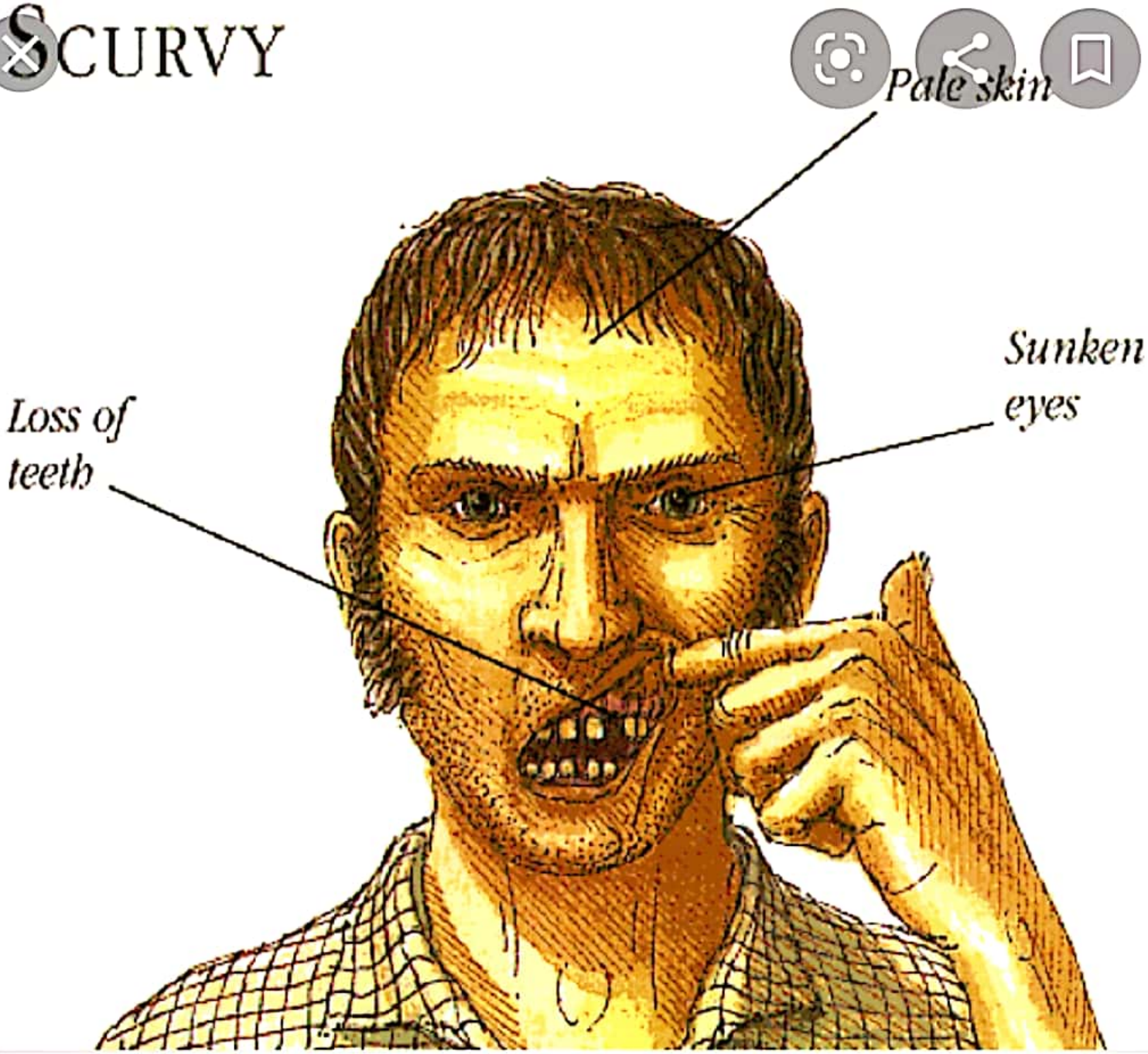
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10 important clinical features in Rickets



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SCURVY



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(9) → Deficiency of Ca^{+2} & vitamin-D in adults is called Osteomalacia

→ X-ray reveals thick & uncalcified osteoid

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(10) → Osteoporosis → bone having pores.
Deficiency of Ca^{+2} in old age
⇓
There is forward bending of vertebral column (Kyphosis)

(11) → Septic arthritis →
- Infection reaching metaphysis.



(12) → Bone marrow biopsy →
Extraction of bone marrow
either from — manubrium sterni
— iliac crest



(13) → Bone tumor → can be benign or malignant

(14) → Rupture of cartilage →
- Medial meniscus of knee mostly affected
- Treated by replacement

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(15) → Tumors of cartilage →

* benign → chondroma
* malignant → chondrosarcomas

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(16) → Cartilage transplantation →
person to person can be done

★ Clinicals of Joints : →

(1) Inter-vertebral disc is secondary cartilaginous joint b/w vertebras

— if nucleus pulposus of disc protrude backward, it can cause damage to spinal nerves

↓

this condition is called Herniation of disc / Disc prolapse ★★★

NOTE → If disc prolapse occurs in Lumbar vertebrae there can be radiating pain in Lower limb

↓

called sciatica

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(2) Dislocation → End of one of the bone out of its socket.

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(3) Sublaxation → End of bone partially leaves socket.

(4) Rheumatic fever → Healing pain in joints. (→ शरीर के जोड़)
* Streptococcal pharyngitis
* Temprory pain
* Above bacteria & toxins can affect mitral valves of heart or the kidneys.

(5) Rheumatoid arthritis →
Inflammatory systemic disease
* Thickened synovial membrane
* Due to chronic inflammatory process there is erosion of bones
⇓
causing deformity in fingers.

(6) Osteo-arthritis →
Degenerative condition of large weight-bearing joints.

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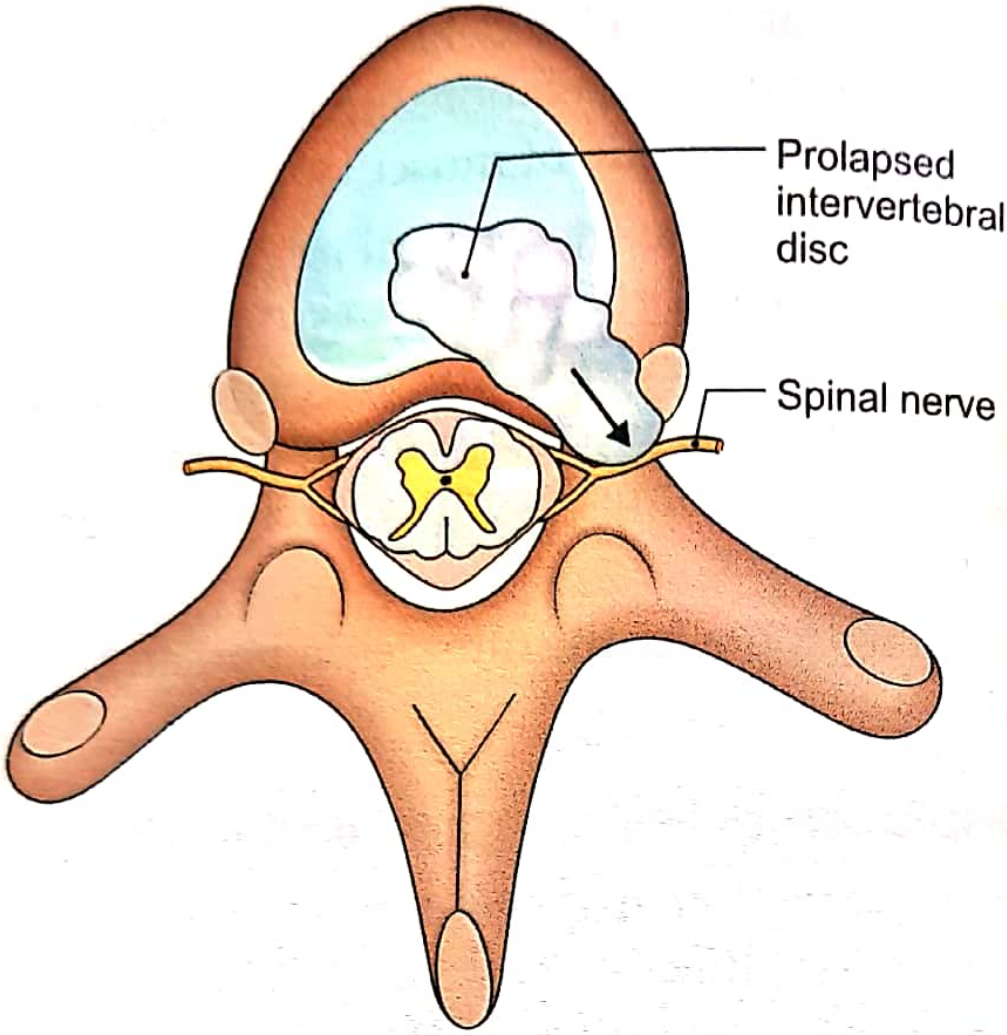


Fig. 3.31: Herniation of the disc

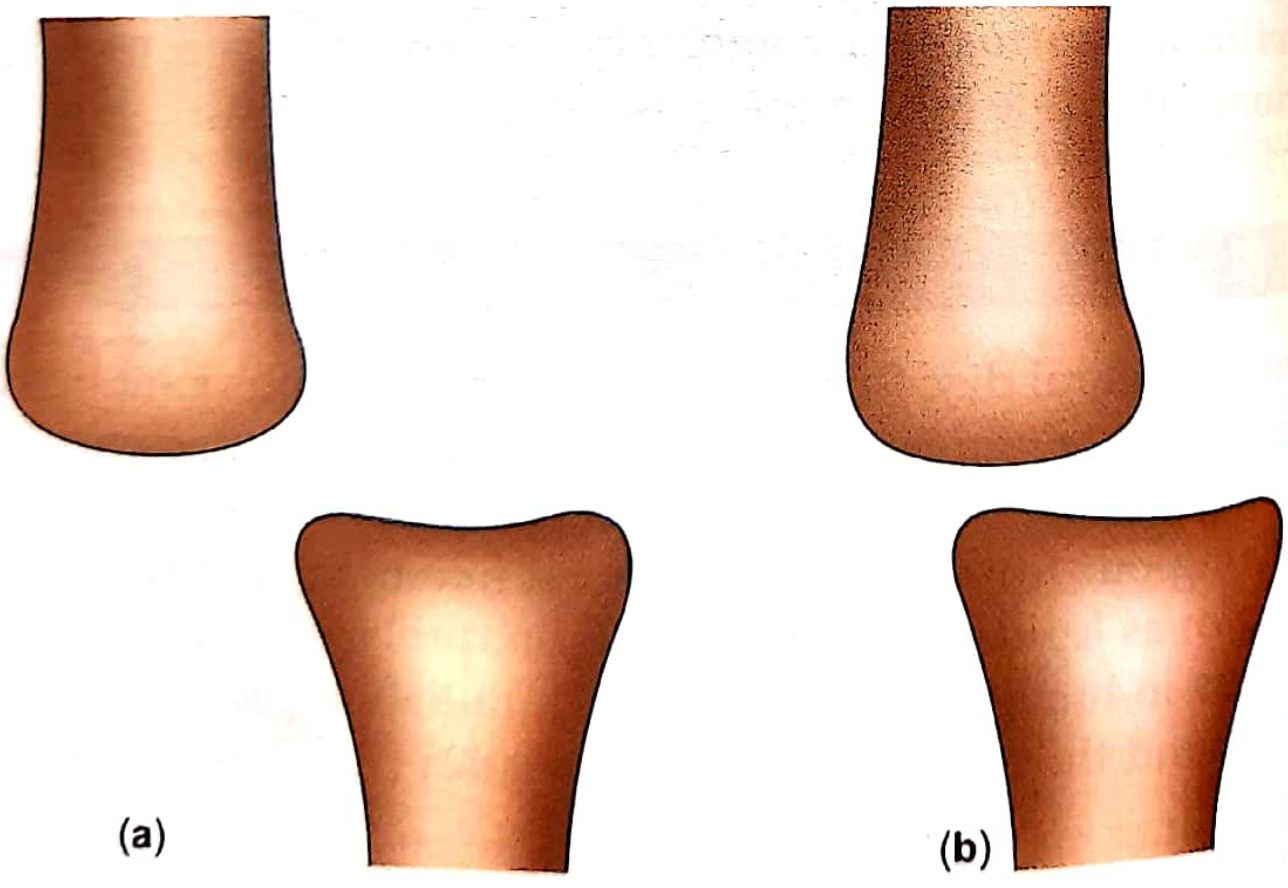


Fig. 3.32: (a) Dislocation, (b) Subluxation

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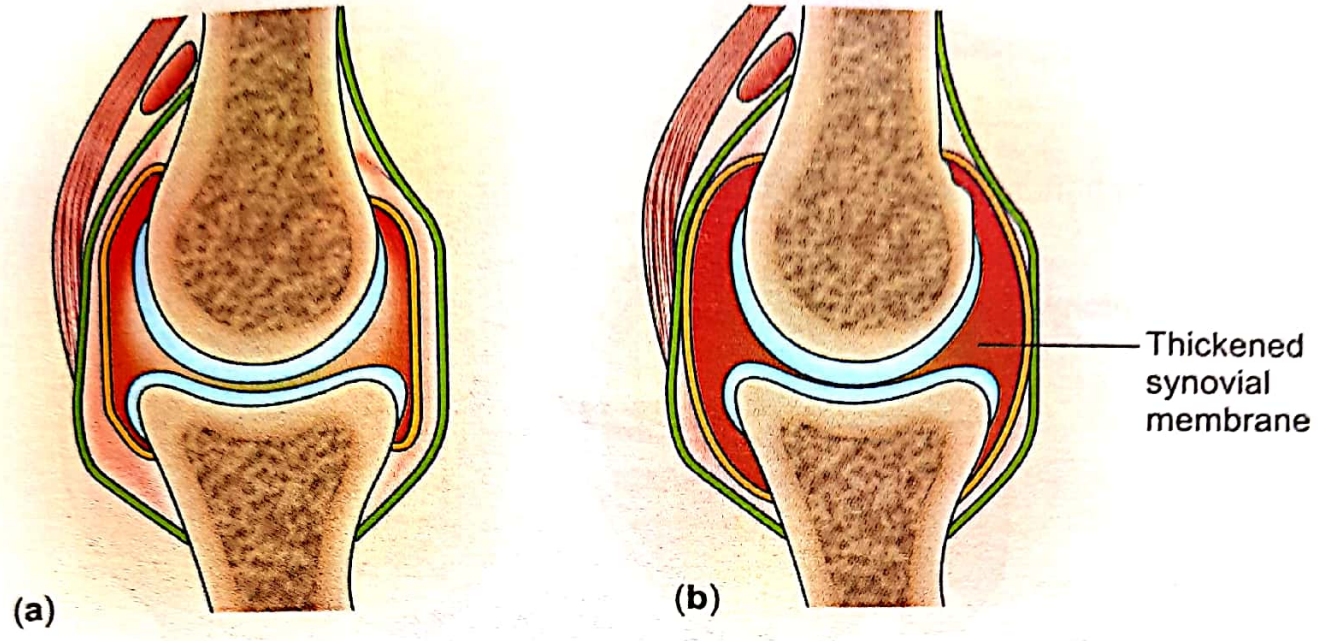


Fig. 3.33: (a) Synovial membrane in a normal joint and (b) thickened in a rheumatoid arthritis patient

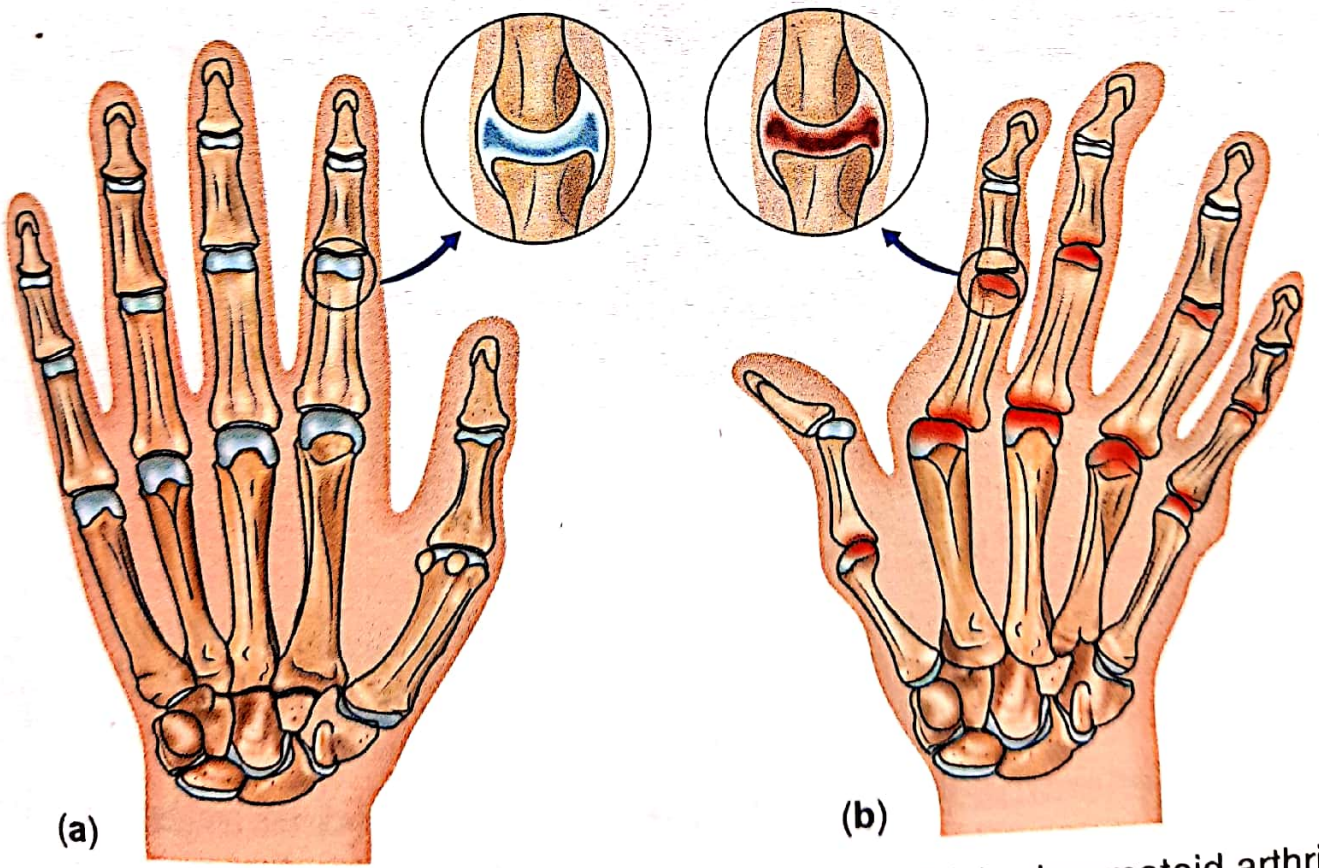


Fig. 3.34: (a) Joints of hand in normal subject and (b) in rheumatoid arthritic hand

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→ Articular cartilage wears out, degenerates
+ there is formation of peripheral osteophytes.

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→ जो होता है bone rub होते को कारण there will be pain

	Osteo-arthritis	Rheumatoid arthritis
① Age & joints	Middle age generally in single large weight-bearing joint-	Any age = multiple small joints of hand & feet
② Disorder	Degenerative	Inflammation
③ Initial damage	Articular cartilage damaged	synovial membrane का inflammation
④ Systemic disease	None	⊕ ve
⑤ ESR	Normal	Raised, Anemia

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⑦ Spondylitis → degenerative changes

→ can occur in cervical spine

↓

leading to narrowed inter-vertebral foramen

↓

causing pressure of spinal Nerves.

⑧ Septic arthritis →

* Metaphysis is actively growing end of bone
* In some joint capsule encloses metaphysis as well
* जो अगर metaphysis में infection है तो it can reach joint capsule

⇓

Septic Arthritis

→ अगर

⑨ Stiffness of joint is related to weather.

V. Imp Viscosity of Synovial fluid ∝ fall in the Temp.

→ इसलिए joint become stiff in cold weather

NOTE → mobility promotes lubrication & decreases stiffness इसलिए with passing day movements are resumed to normal

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(10) Neuro-pathic joint → result of complete denervation

→ so, all reflexes are eliminated & joint is open to mechanical damage

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→ shows → (a) swelling
(b) excessive mobility
(c) bony destruction

→ Happens in → (a) leprosy
(b) Tabes dorsalis
(c) Syringo-myelia

cyst = syrinx

formed in spinal chord

Overtime destroys spinal chord.

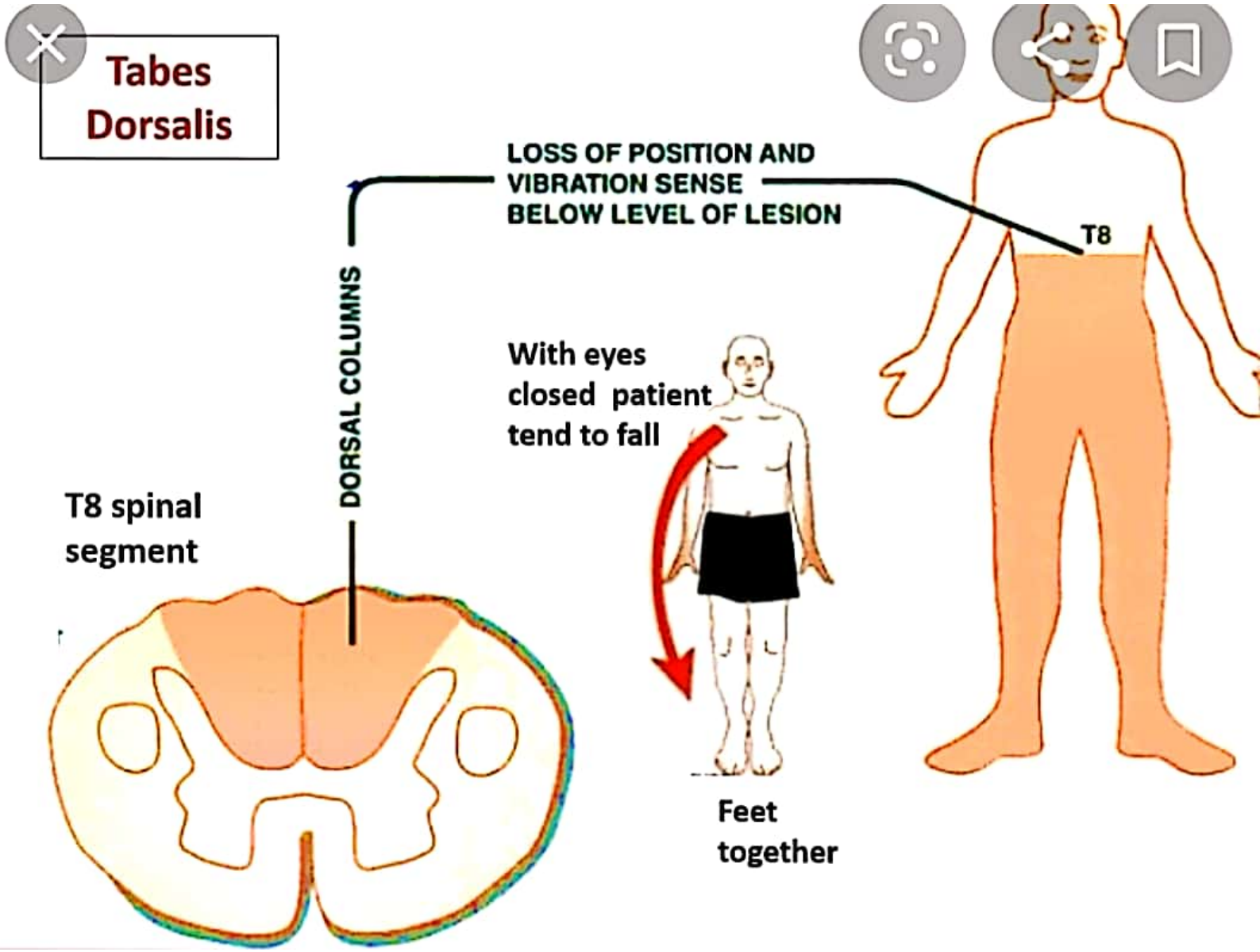
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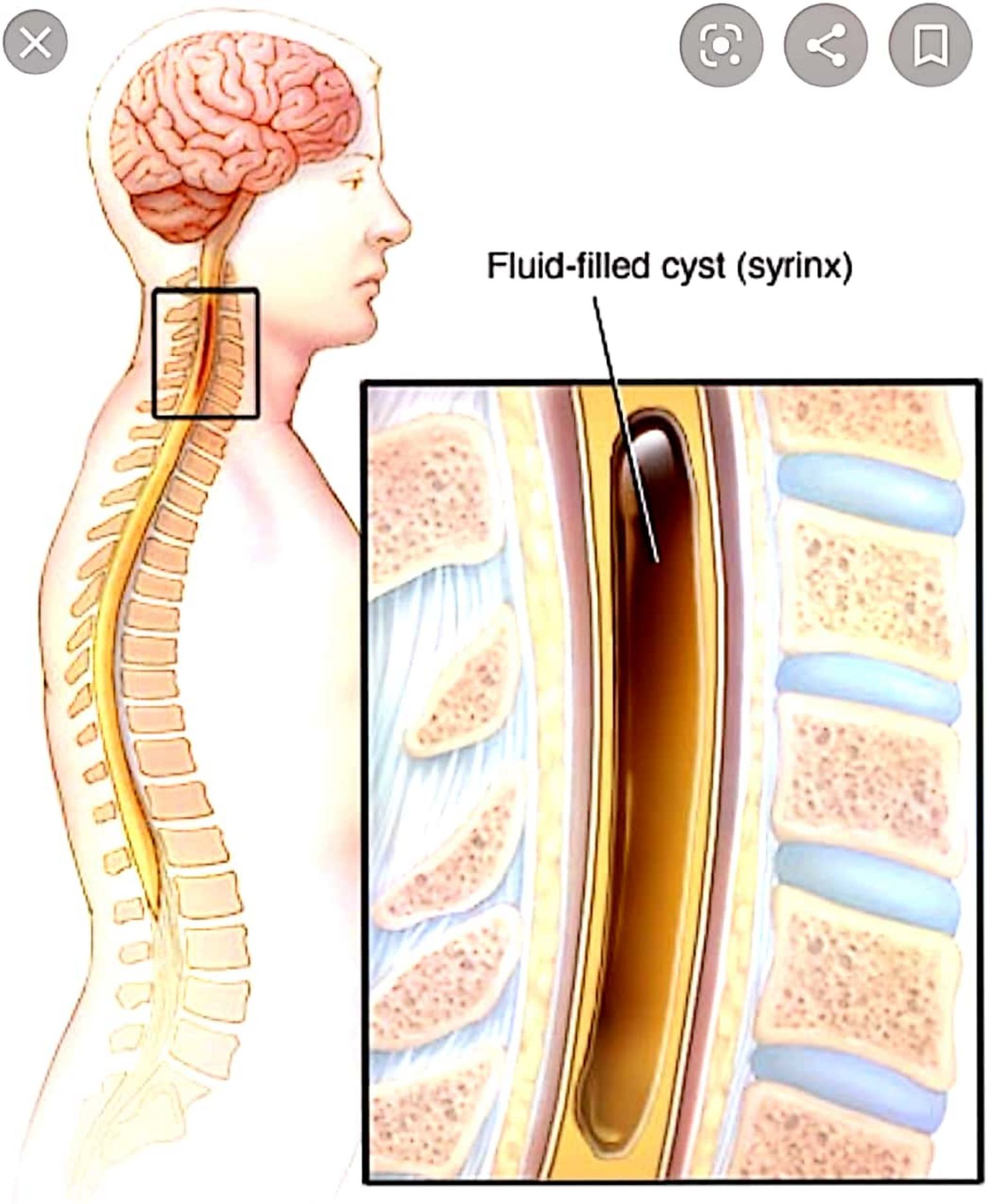
Symptoms of LEPROSY

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Tabes Dorsalis



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★ Muscles :→

→ (mus = mouse) as most of muscles resemble like mouse
Tendon as its tail

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Origin → Generally at part which remains fixed
Insertion → Generally at which moves
} However can be interchange

NOTE → In limb muscles origin is usually proximal to insertion

* → Muscle has two parts
(A) Fleshy part = called belly
(B) Fibrous part = noncontractile & inelastic
→ Rope like → Tendon
→ Flattened → Aponeurosis

* Muscle is supplied via capillaries

Arrangement → Muscle → fasciculi → Fibre → Myofibril → Myofilaments.

* ★
A-band → H-zone
I-band → Z-line

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Table 4.1: Types of muscles

	Striated/skeletal	*Non-striated/smooth*	*Cardiac*
1.	Striated muscles are present in the limbs, body wall, tongue, pharynx and beginning of oesophagus (Fig. 4.1)	Oesophagus (distal part), urogenital tract, urinary bladder, blood vessels, iris of eye, arrector pili muscle of hair (Fig. 4.2)	Wall of heart (Fig. 4.3)
2.	Long and cylindrical	Spindle shaped	Short and cylindrical
3.	Fibres unbranched	Fibres unbranched	Fibres branched
4.	Multinucleated	Uninucleated	Uninucleated
5.	Bounded by sarcolemma	Bounded by plasma-lemma	Bounded by plasma-lemma
6.	Light and dark bands present	Light and dark bands absent	Faint light and dark bands present
7.	No intercalated disc	No intercalated discs	Intercalated disc present and a characteristic feature
8.	Nerve supply from cranial nervous system	Nerve supply from autonomic nervous system	Nerve supply from autonomic nervous system
9.	Blood supply is abundant	Blood supply is scanty	Blood supply is abundant
10.	Very rapid contraction	Slow contraction	Rapid contractions
11.	They soon get fatigued	They do not get fatigued	They never get fatigued
12.	Voluntary	Involuntary	Involuntary

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Slow fibres	Fast fibres
① Red in colour	① White in colour
② Large amount of Myoglobin	② Small amount of Myoglobin
③ Rich in mitochondria	③ Less no. of mitochondria
④ Aerobic	④ Anaerobic
⑤ Low fatigue	⑤ Easily fatigued

★ Naming the muscles : →

→ on the basis of

① Shape : →

(1) Deltoid → Triangular

(2) Quadratus = Quadrangular

Ex → Quadratus femoris

(3) Rhomboid = Diamond shaped

Ex → Rhomboid major, minor

(4) Teres = Round

Ex → Teres major, minor

(5) Gracilis = पतला

Ex → Gracilis

(6) Lumbricals = Worm like

Ex → Lumbricals of palm

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(1) Rectus = Straight
Ex → Rectus abdominis

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(2) Size →

(1) Major - big
Ex → Pectoralis major

(2) Minor - small
Ex → Pectoralis minor

(3) Longus - long
Ex - Adductor longus

(4) Brevis - small
Ex - Adductor pollicis brevis

(5) Latissimus - broadest
Ex - Latissimus dorsi

(6) Longissimus - Longest
Ex - Longissimus thoracis

(3) Number of heads →

(1) Biceps - two heads

(2) Triceps - three heads

(3) Quadriceps - four heads
Ex → Quadriceps femoris

(4) Diagastric - two bellies
Ex → Ant. & Post. bellies.

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④ Attachment →

(1) Sterno-cledo-mastoid →
from sternum & clavicle → to mastoid process

(2) Brachialis → 2/3 brachium (Humerus to Ulna)

(3) Coraco-brachialis → from coracoid process to Arm

⑤ Depth →

(1) Superficialis (superficial) → F.D.S
(2) Profundus (deep) → F.D.P
(3) Externus (external) → E. Obligue
(4) Internus (Internal) → 2. Obligue

⑥ Position

(1) Dorsi → of the back.
Ex → Latissimus dorsi
(2) Brachii → of the Arm
Ex → biceps brachii
(3) femoris → of the thigh
Ex → rectus femoris.
(4) Oris → of the mouth
Ex → orbicularis oris
(5) Oculi → of the eye
Ex → orbicularis oculi

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⑦ Action →

① Extensor → ↑ the ∠ b/w two

② flexor → ↓ the ∠ b/w two

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③ Abductor → take away

④ Adductor → towards midline

⑤ Levator → to elevate

⑥ Depressor → to pull down

⑦ Supinator → turning palm Anteriorly

⑧ Pronator → turning palm Posteriorly

⑨ Constrictor → to narrow

⑩ Dilator → to dilate

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→H
→Z

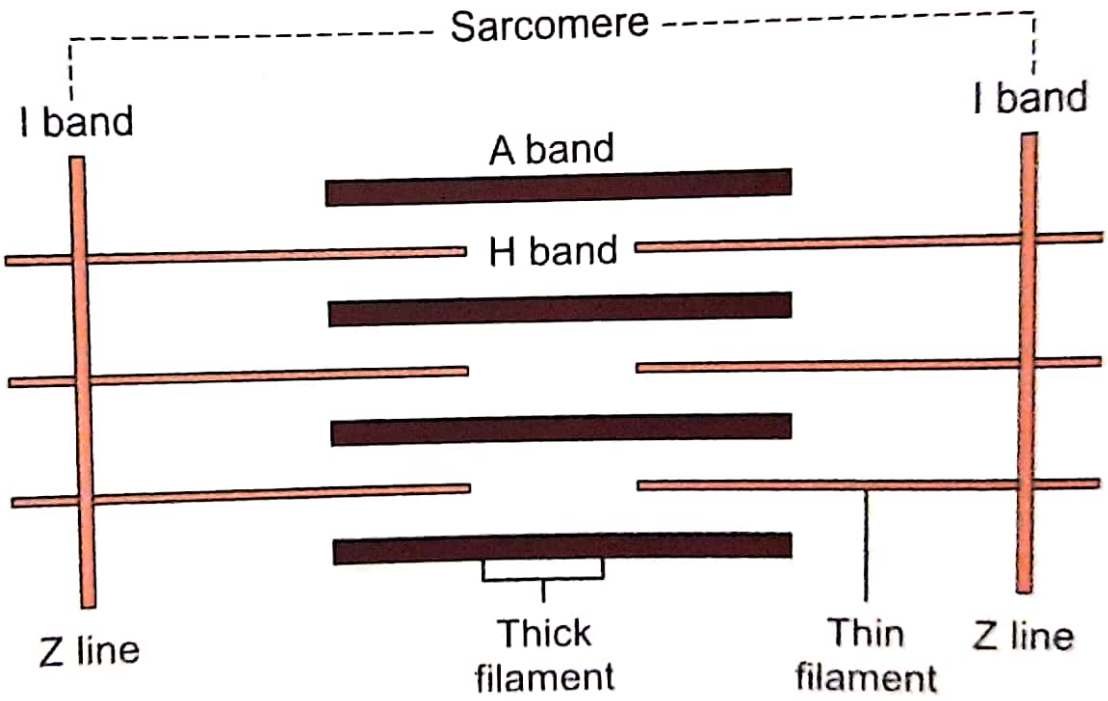


Fig. 4.5: Myofibrils in skeletal muscle

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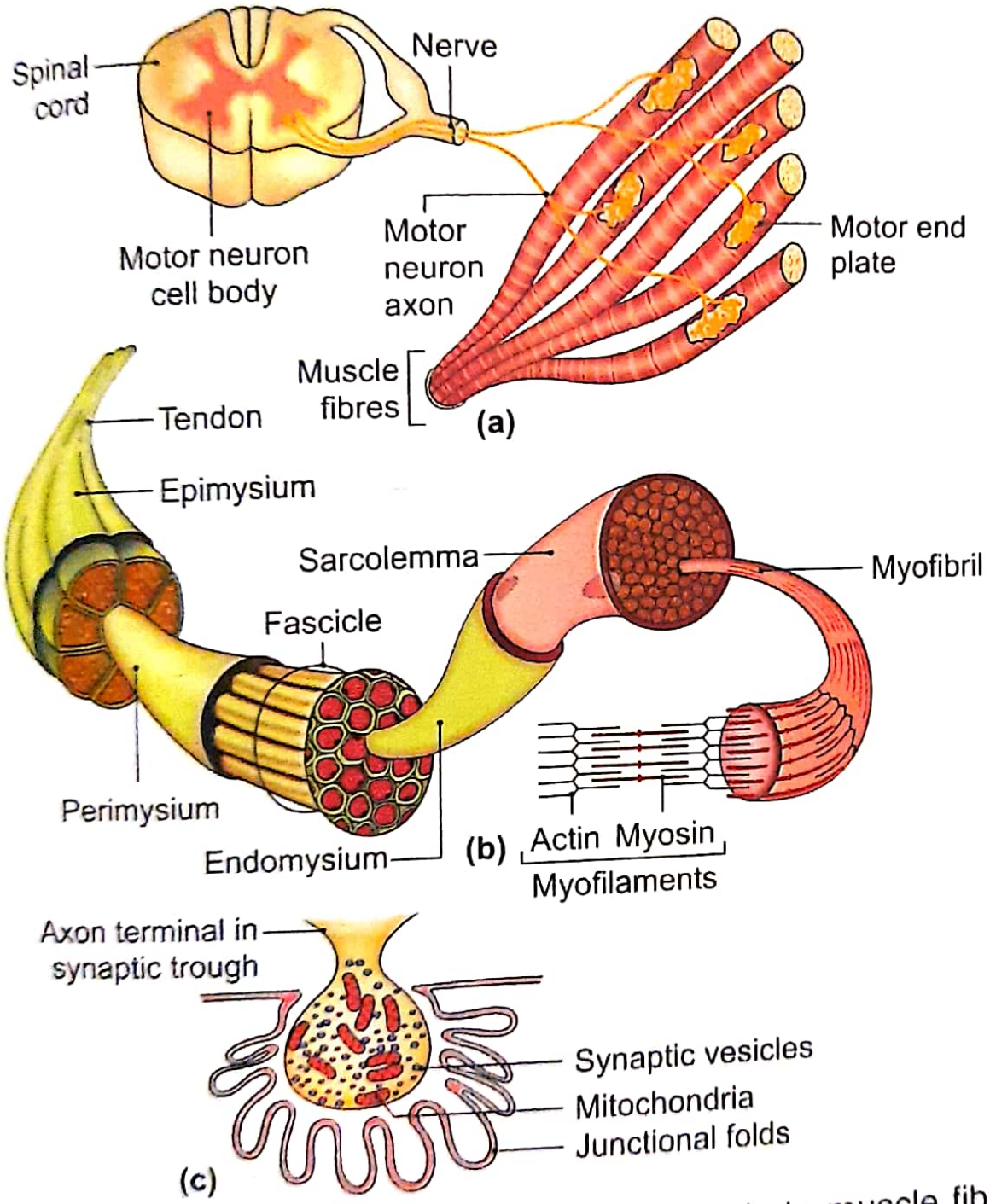


Fig. 4.6: Supporting tissue of a muscle. Nerve supply to muscle fibres also shown (a) nerve supply to muscle fibres; (b) supporting tissue of a muscle; (c) motor end plate

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★ Clinical of- Muscles →

① Paralysis

→ Loss of motor power (power of movement)

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→ Inability of muscles to contract-

Q. How ?

→ (1) either damage to motor neuron pathways (upper & lower motor neuron)

(2) or by inherent disease of muscle (myopathy)

★★★ V. Imp

(1) Upper motor neuron damage = Spastic Paralysis

⇓

have exaggerated tendon jerk.

(2) Lower motor neuron damage = Flaccid Paralysis

⇓

Loss of tendon jerk.

★ V. Imp

→ Lower motor neuron paralysis occurs in Poliomyelitis.

② Muscle spasm →

→ Quite painful

→ Locally, commonly caused by muscle pull.

→ In order to relieve its pain the muscle should be relaxed by appropriate treatment-

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→ Generalized spasm occurs in
= (1) Tetanus
= (2) Epilepsy.

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(3) Disuse atrophy & hypertrophy →
→ muscle not used for long time become thin & weak → Disuse atrophy.

→ Adequate or excessive use of particular muscle can cause → better development, or hypertrophy

★ NOTE → Muscular 'wasting' → [reduction in size] ★ is a featur of Lower motor neuron paralysis

(4) Regeneration of skeletal muscle →
→ skeletal muscle capable of limited regeneration,
→ if large damage → does not occur
→ missing muscle is replaced by connective tissue

(5) Hyper-plasia → increased size with ↑ed cell no. → occurs in uterus during pregnancy.

NOTE → Tone is resitance to touch.

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(6) Myasthenia gravis →

→ Autoimmune disorder

→ The nerve impulse transmission to muscle fibre is blocked

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Reason → Antibodies bind Acetylcholine receptors

★★★

* This causes - extensive
- progressive
muscle weakness although muscle is normal

NOTE → Extraocular and muscle of eyelid are affected 1st

→ followed by those of neck & limb.

(7) Poly-myositis →

→ characterised by inflammation of muscle fibres

→ It starts when WBC's invade muscles

* → Muscles close to trunk are mostly affected result in severe weakness.

→ Polymyositis + skin rash → Dermamyositis

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⑧ Fibrillation →

→ Abnormal contraction of cardiac muscle.

→ दो तरह के cardiac chambers

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do not contract as a whole resulting in disruption of pumping action.

~ Atrial → Rapid & uncoordinated contraction of Atria.

↓

Ineffective pumping & abnormal contraction of AV-node

~ Ventricular → very rapid & disorganised contraction of ventricle.

↓

leading to disruption of ventricular function.

⑨ Angina-pectoris →

- Episode of chest pain due to temporary ischaemia of cardiac muscle
- relieved by = Rest & Nitrites

⑩ Myocardial Ischemia →

→ Ischemia due to blockage of more than 1 artery or main artery causes necrosis of cardiac muscle.

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V. imp points →

① Cardiac muscle → least
Smooth muscle → Intermediate
Skeletal muscle → Maximum
Amount and weight

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② some muscle are vestigial
Ex → (i) muscles of Auricle
(ii) Palmaris brevis of palm

③ some tendons & ligaments are divorced part of muscle.
— Palmans aponeurosis = part of palmaris Longus
— Plantar aponeurosis = plantaris
— Tibial collateral lig. = Adductor magnus
— Sacro-tuberous lig. = degenerated tendon of long head of biceps femoris.
— Sacrospinous lig. = Degenerated part of coccygeus
— Long plantar lig. = part of peroneus longus.

★★ Longest muscle → Sartorius

★★ Longest tendon → Tendocalcaneus

★★ Soleus = peripheral heart contain many venous sinuses.

★★ Largest muscle → Gluteus maximus.

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④ Red muscle fibre is used in marathon

★★ Richest Nerve supply → Extraocular muscles

Reason → due to small motor units.

⑤ Smooth muscle has max. regeneration capacity

⑥ Cardiac muscle do not regenerate.

★★ Muscle used for Intramuscular injection → (i) Deltoid
(ii) Gluteus medius
(iii) Vastus lateralis.

⑦ — Muscle fibre supported by → Endomysium
— Muscle fasciculus → Perimysium
— whole muscle. → Epimysium

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✱ → Circulatory system →

① BP → systolic BP
→ Diastolic BP
→ pulse pressure

② Haemorrhage → bleeding results from rupture of blood vessel

Arterial — spurting of blood (धारदार)

Venous — pooling of blood (ठहरा)

③ Vascular catastrophies →
* Thrombosis → intravascular clot
* Embolism → rupture & movement of clot
* Haemorrhage

④ Arterio-sclerosis →
- In old age arteries become stiff.
- causes wide decrease in blood supply to tissue & rised systolic pressure

⑤ Arteritis & phlebitis →
↓ Inflammation of Artery
↳ Inflammation of vein

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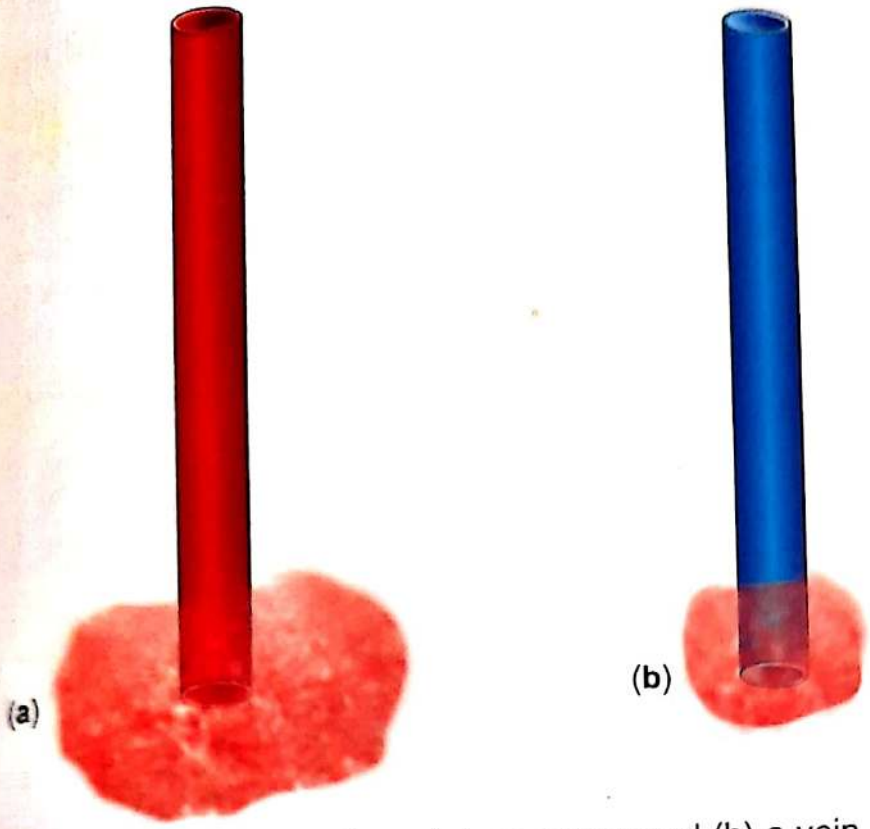


Fig. 5.24: Bleeding from (a) an artery and (b) a vein

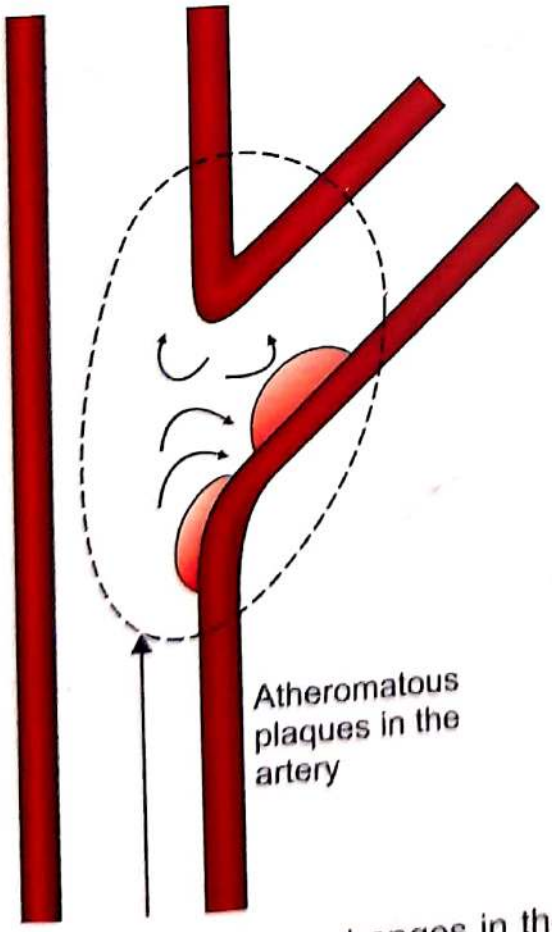


Fig. 5.25: Atheromatous changes in the artery

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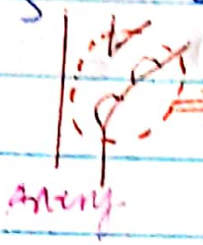
⑥ Atheroma →

→ Patchy changes in tunica intima of Artery due to

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cholesterol, other lipid compounds.

→ The change is called Athero-matous plaque



⑦ Coronary artery blockage →

→ Graft is generally longest vein, great saphenous vein

or

Anterior thoracic Artery.

⑧ Aneurysm →

- swelling or dilation of blood vessel where part of Artery inflate like a balloon.

- वहाँ पर wall is weaker & thinner इसलिए it is risky तभी फट सकती है।

⑨ Coronary Angiogram →

Angiography of coronary Artery.

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arteries
coronary angiogram

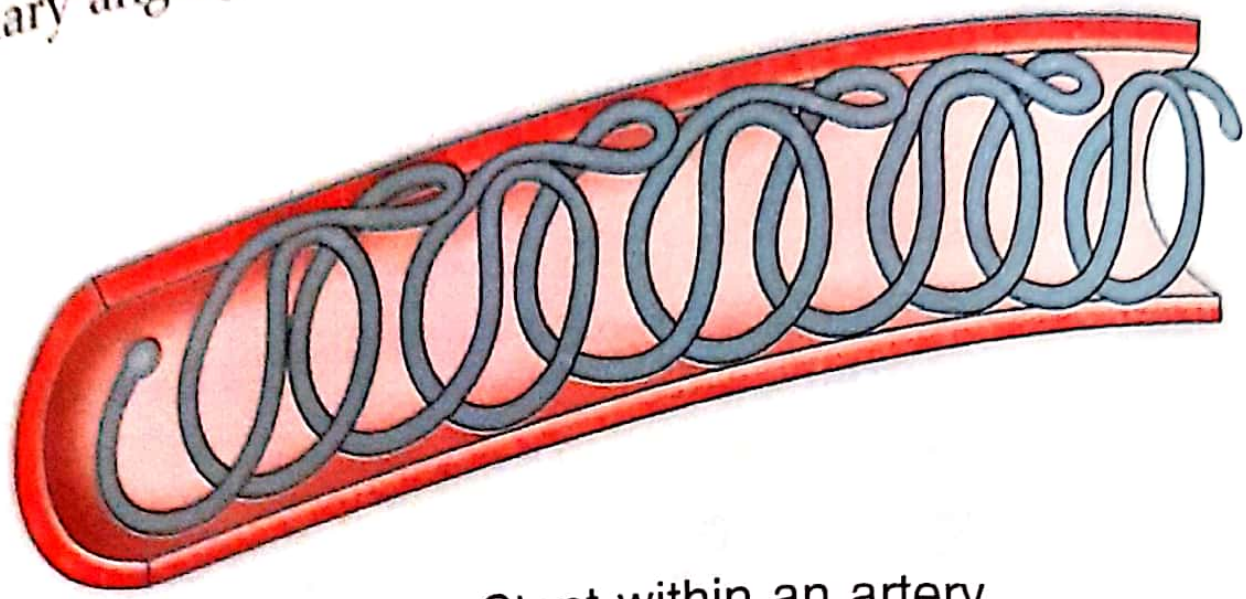


Fig. 5.26: Stent within an artery

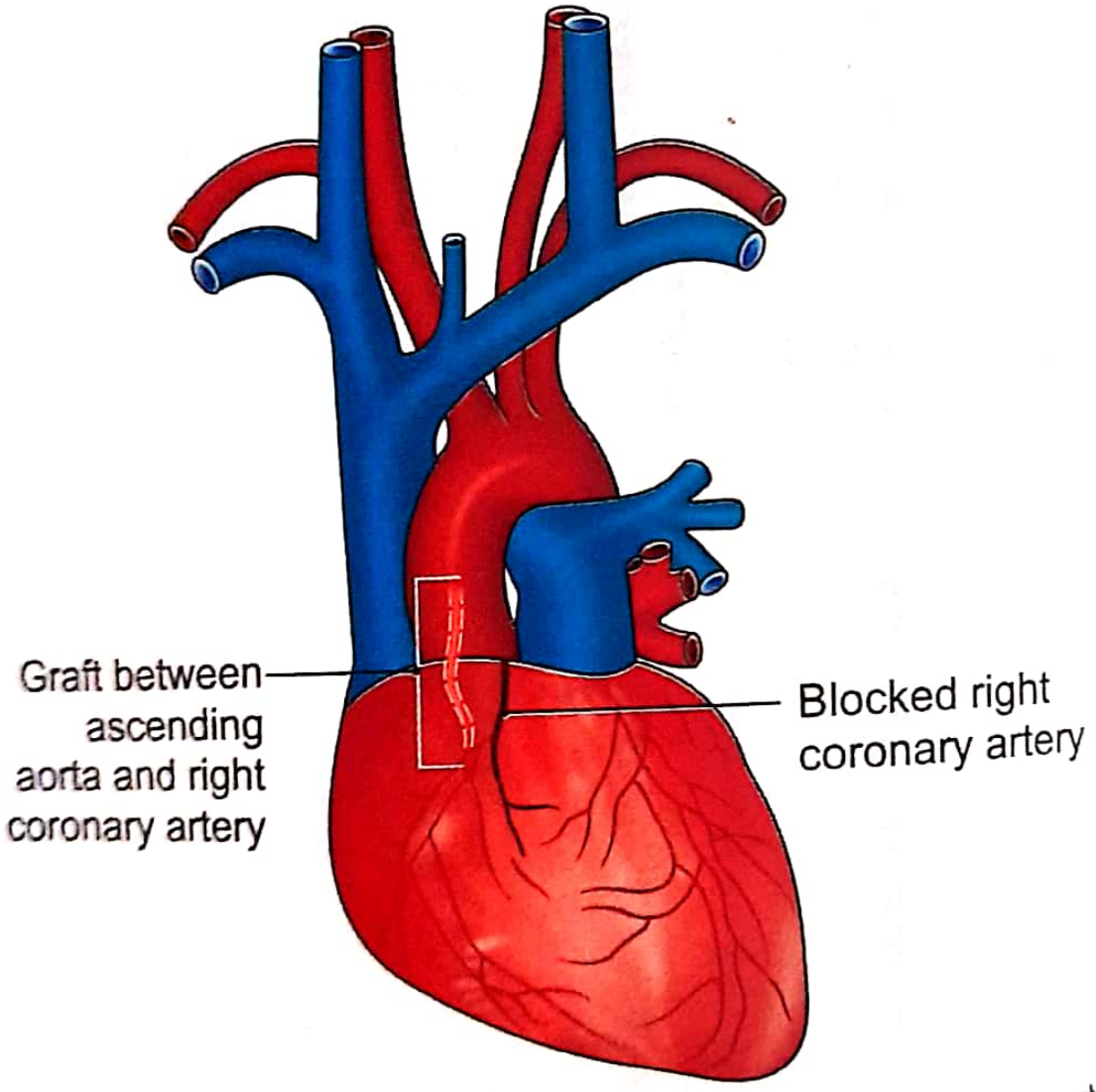


Fig. 5.27: Graft for bypassing the blocked coronary artery

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(10) Buerger's disease

→ Thrombo-angiitis Obliterans

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* very painful
* Inflammation of smaller peripheral arteries of legs.
* Victim → young person & heavy smoker.

(11) Raynaud's phenomenon

→ pallor of fingers
→ due to constriction of smaller arteries & arterioles in response to cold.

(12) Acute phlebo-thrombosis →

→ veins of lower limbs are affected
→ Due to lack of movement of leg
↓
there is thrombus formation with mild inflammation
↓
सीधा embolism हो सकता है & can block any other artery.

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Healthism

Visit

Buerger's Disease | Healthism

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5 Causes for Raynaud's Disease or Raynaud's Phenomenon



1. Diseases of the arteries (atherosclerosis)
2. Drugs that cause narrowing of the arteries (amphetamines, some beta-blockers, some cancer drugs, some migraine medications)
3. Certain autoimmune conditions (SLE-lupus, scleroderma, sjogrens, RA)
4. Smoking
5. Repeated injury or usage (i.e., typing, piano, heavy use of hand tools)

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(13) Varicose veins →

→ जब vein पे लगातार pressure हो Page No. Date
for longer period of time

↓

there is Atropy of muscle and elastic tissue with fibrous replacement

↓

|| becomes ≀≀

Treatment → putting leg on stool higher than trunk जिससे venous return comes to normal

→ varicose vein may occur at-
- ✓ lower end of Oesophagus
- ✓ Anal canal.

(14) At times → Parenteral nutrition → given through right subclavian vein.

(15) blood vessels in retina can be examined by
→ Opthalmoscope

★ → In Diabetes & Hypertension.

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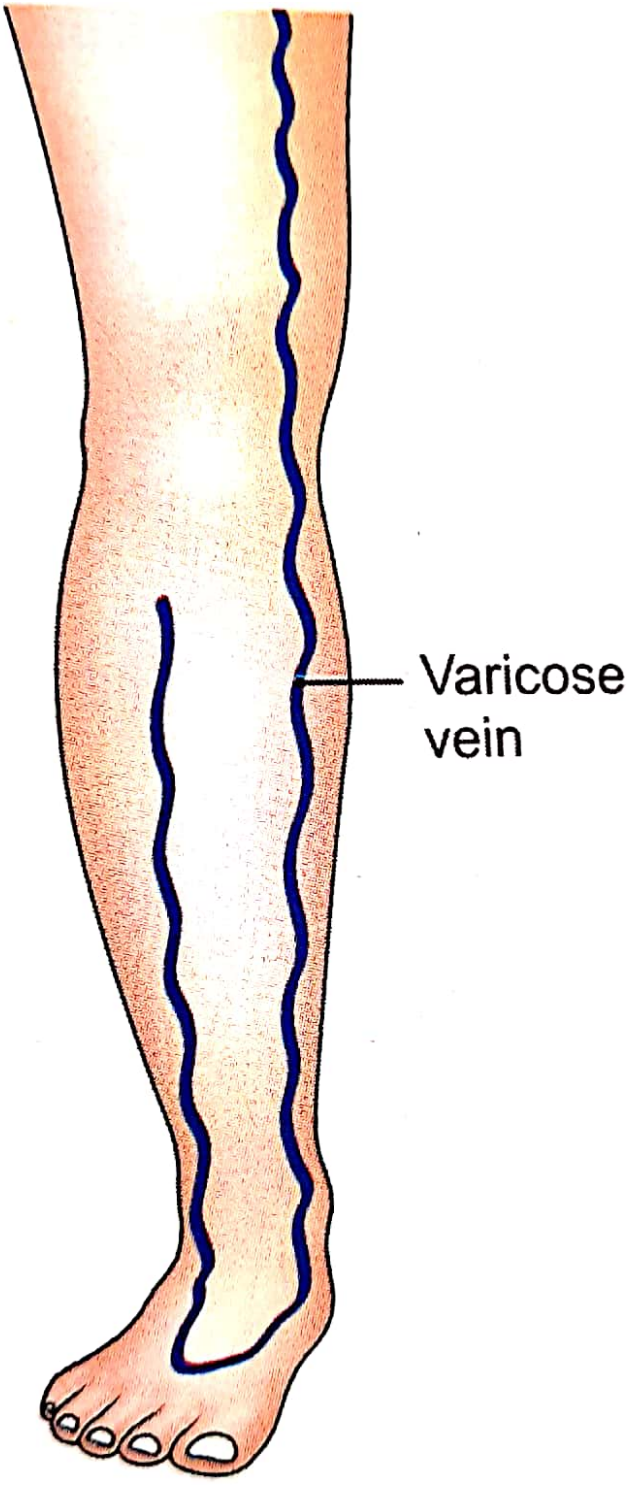


Fig. 5.28: Varicose veins in lower limb

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NOTE → Arteries have thickest → Tunica media

→ Veins have thickest → Tunica Adventitia

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★★ Some Palpable Arteries → Radial, common carotid, facial, superficial Temporal

V.Imp ★ Lymphatic system →

ये एक drainage system है जो veins के साथ-साथ चलता है।

→ ऊतक से है जब artery का blood आता है supplying organ through capillaries the blood is sent back into veins.

ये होते साथ tissue fluid बनता है which is absorbed by → venous end of capillaries
→ & post-capillary venules

→ Along with it about (10-20%) is absorbed in lymphatics.

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to the arteries, but

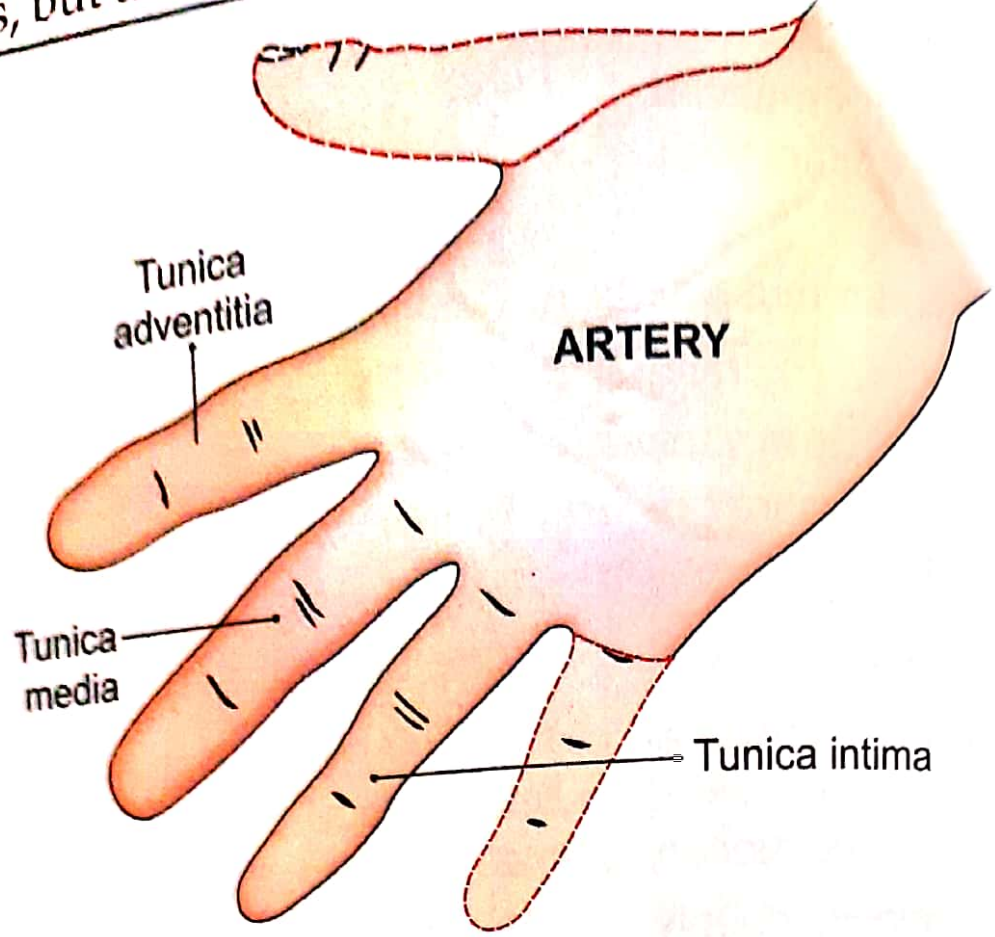


Fig. 5.9: Artery with thick tunica media

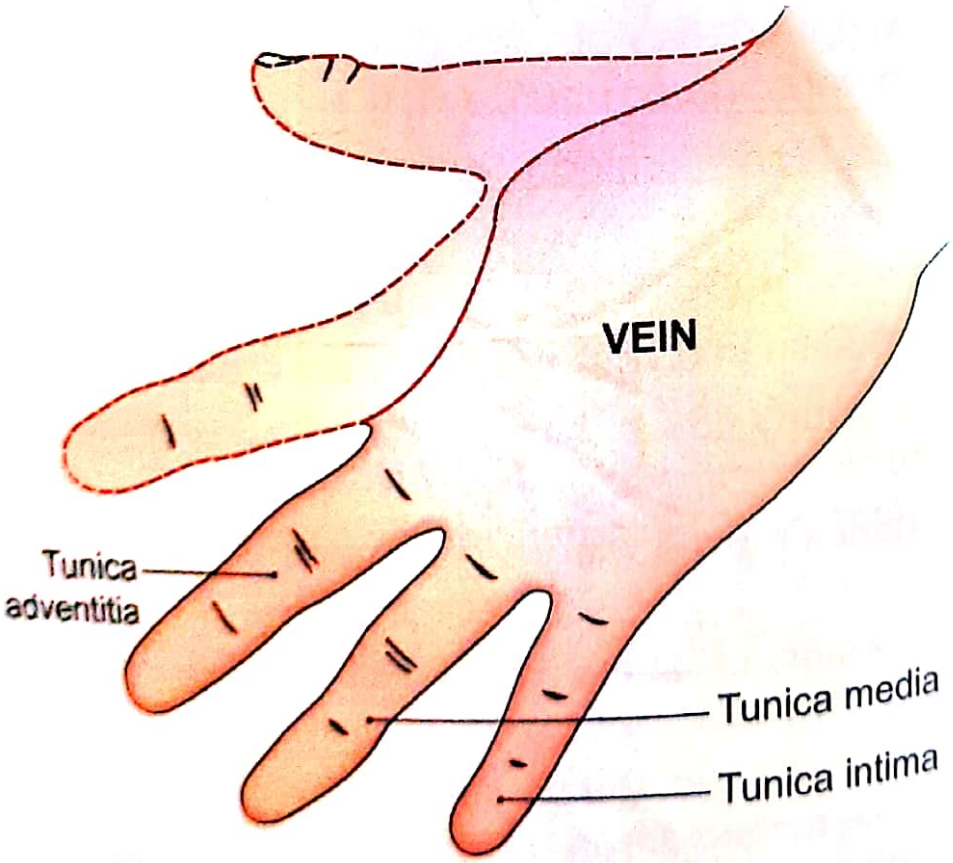


Fig. 5.10: Vein with thick tunica adventitia

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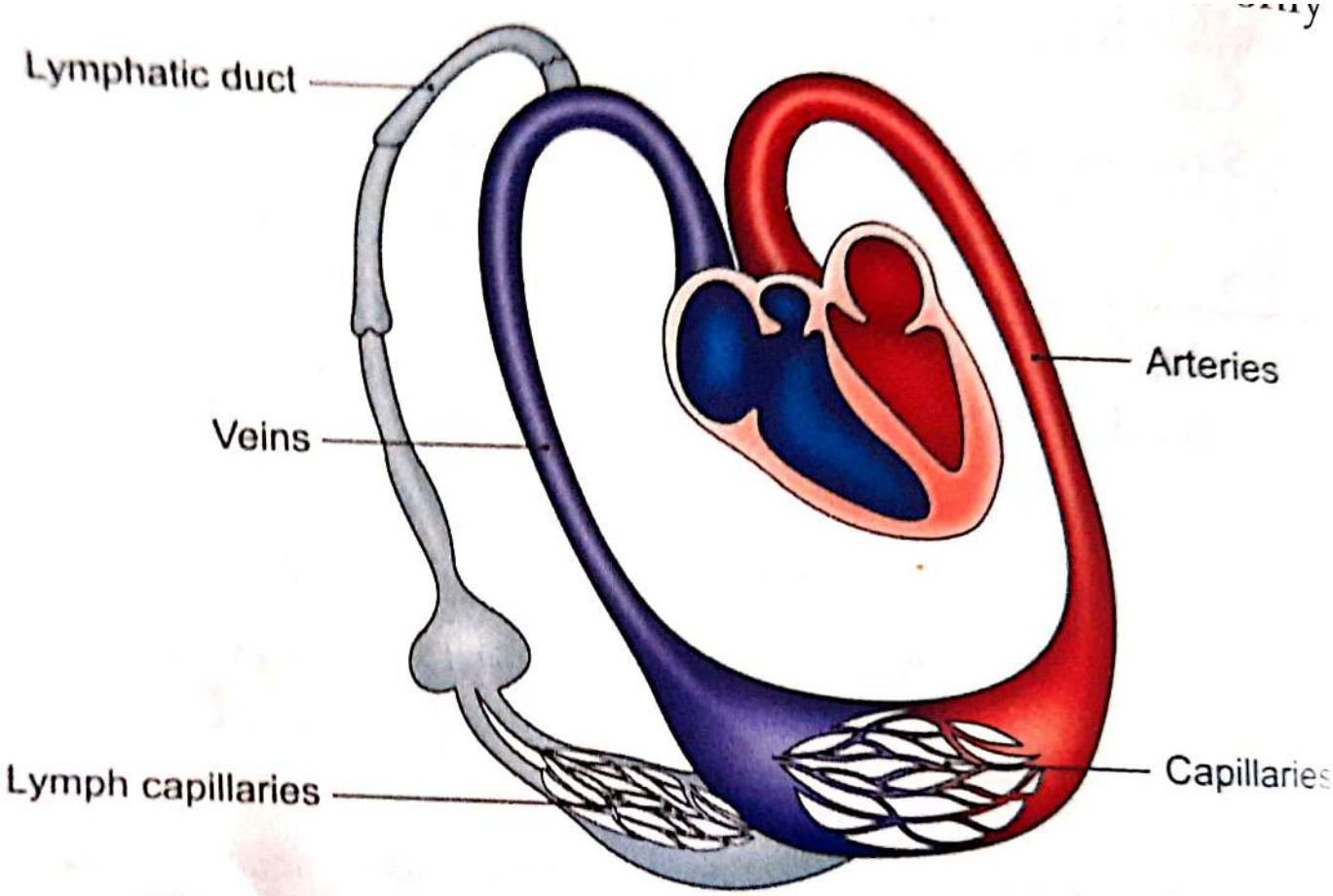


Fig. 6.1: Beginning and termination of lymph vessels

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